

Surface of Silicon



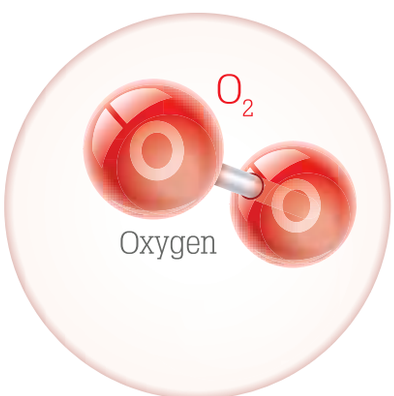
The most abundant atom in the universe is the hydrogen atom. Nearly 74% of the atoms in the universe are hydrogen atoms. However, three most abundant atoms on the Earth are iron, oxygen and silicon.

3.2 Molecules

When an atom combines with another atom (or atoms) and forms a compound, it is called as molecule. *A molecule is made up of two or more atoms chemically combined.*



Oxygen gas in the air that we breathe is made up of two oxygen atoms chemically combined.

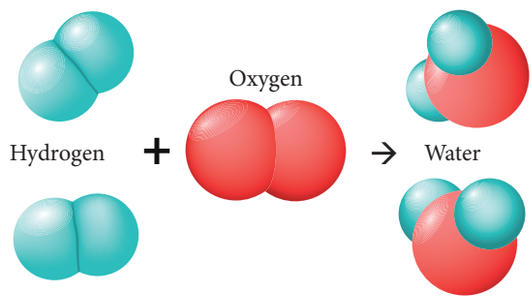


Ozone is a substance that is made up of three oxygen atoms chemically combined.



An atom of oxygen (O) and two atoms of hydrogen (H) combine to form a molecule of water (H_2O).

$$2H_2 + O_2 \rightarrow 2H_2O$$



$$\mathbf{2H_2 + O_2 \rightarrow 2H_2O}$$

Formation of water molecule

Molecules exhibit the properties of matter and also have individual existence. A molecule can be formed by the same or different kinds of atoms.

Molecules can be classified as below.

- A molecule which contains only one atom is called monatomic molecule (Inert gases).
- A molecule which contains two atoms is called diatomic molecule (Oxygen, Nitric oxide, Hydrogen, etc.).
- A molecule containing three atoms is called a triatomic molecule (Ozone, Sulphur dioxide, Carbon dioxide, etc.).
- A molecule containing more than three atoms is known as polyatomic molecule (Phosphate, Sulphur, etc.).

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